

```
$ /opt/qemu-0.15/bin/qemu-img create ubuntu2.img 2000M
$ /opt/qemu-0.15/bin/qemu -hda ubuntu2.img -cdrom /home/
anton/ISO/ubuntu-10.10-server-1386.iso -M pc -m 768 -vga
cirrus -no-acpi
```

Similarly, you can use other OSs, for example, Windows 2003 EE:

```
$ /opt/qemu-0.15/bin/qemu -hda windows2003.img -cdrom /home/
anton/ISO/win2003_sp2.iso -M pc -m 1024 -vga cirrus
```



A small tip: If you can't press the keyboard Ctrl+Alt+Del, then go to the QEMU console with Ctrl+Alt+2. Here, type `sendkey ctrl-alt-delete` and press Enter. The OS in the VM will receive the keyboard combination.

Disk management

Let's imagine that you have a new drive to be used for data storage. It could be a special, dedicated database storage pool. This drive should be initialised first, partitioned with a file *format* that is used by OI. You can use the *format* command (it can be used in interactive mode):

```
root@openindiana:/home/anton# format
Searching for disks...
Failed to inquiry this logical diskdone
```

```
AVAILABLE DISK SELECTIONS:

 0. c1d0 < cyl 2868 alt 2 hd 255 sec 63>
    /pci@0,0/pci-ide@1,1/ide@0/cmdk@0,0
 1. c2d1 <QEMU HAR-QM0000-0001-2.00GB>
    /pci@0,0/pci-ide@1,1/ide@1/cmdk@1,0
```

Our main OS drive is shown as number 0 (*c1d0*). Choose the newly inserted drive which is identified as `"c2d1 <QEMU HAR-QM0000-0001-2.00GB>".` We can format the whole drive with the *format* option in the menu (not shown), or assign a label using *volname*. The last step is to add the drive to the ZFS pool—and the new storage space is ready to use:

```
root@openindiana:/home/anton# zpool create pool c2d1
```

You can list the available pools and see that both pools (i.e., disks) are online:

```
root@openindiana:/home/anton# zpool list
NAME      SIZE  ALLOC   FREE   CAP  DEDUP  HEALTH  ALTROOT
pool      1.98G  124K    1.98G   0%   1.00x  ONLINE  -
rpool     21.9G  9.17G   12.7G   41%   1.00x  ONLINE  -
```

Zones management

In addition to VirtualBox and QEMU virtualisation, OI (and OpenSolaris) offers fairly well-developed virtualisation within the OS, called Solaris Zones [2]. Unfortunately, OI does not yet support zones that can run Linux (though it was available in the previous OpenSolaris version). OI supports mere native zones—when the system runs zones (or containers) with OI inside them. Let's explore how to create them. Begin by using the following code:

```
root@openindiana:/home/anton# zonecfg -z NewZone
NewZone: No such zone configured
Use 'create' to begin configuring a new zone.
zonecfg:NewZone> create
zonecfg:NewZone> add net
zonecfg:NewZone:net> set physical = rtls0
zonecfg:NewZone:net> set address = 192.168.254.200
zonecfg:NewZone:net> end
zonecfg:NewZone> set zonepath=/pool/NewZone
zonecfg:NewZone> commit
zonecfg:NewZone> exit
```

You have created a new zone. It is stored under */pool/NewZone* on the disk you just attached and initialised. For a zone to be complete, you must install the bare OI system into this area:

```
root@openindiana:/home/anton# zoneadm -z NewZone install
A ZFS file system has been created for this zone.
Publisher: Using openindiana.org (http://pkg.openindiana.
org/dev-11/ ).
Image: Preparing at /pool/NewZone/root.
Refreshing catalog 1/1 openindiana.org
Caching catalogs ...
Cache: Using /var/pkg/download.
Sanity Check: Looking for 'entire' incorporation.
Installing: Core System (output follows)
```

```
Packages to install: 1
Create boot environment: No
```

```
[...snip...]
```

```
Postinstall: Copying SMF seed repository ... done.
Postinstall: Applying workarounds.
Done: Installation completed in 771.344 seconds.
```

```
Next Steps: Boot the zone, then log into the zone console
(zlogin -C)

to complete the configuration process.
```

The system is installed into an isolated container. Let's run it!